DOCUMENT APPROVAL SYSTEM

Why Is Meta Interested in Adopting AV1 for RTC?

The motivation for switching to a more advanced video codec is straightforward — it delivers the same visual quality while using much less bandwidth. In offline tests, we observed at least a 20% bitrate reduction with AV1 compared with H.264/AVC under our product settings on low-end and mid-range devices. If devices can accommodate higher encoding complexity, the bitrate reductions are even greater. For real-time video calls, this means people on slower or limited networks can enjoy significantly better video quality. This is important to our users because, to meet low-latency requirements, the RTC product must handle bitrate fluctuations. In real-world networks — especially in emerging markets — video bitrates for RTC products typically range from 10 kbps to 400 kbps. Maintaining good video quality below 100 kbps remains challenging.

To evaluate the user experience across codecs, we enabled AV1 in the Messenger app and conducted a side-by-side comparison using two Android phones. In the examples below, AV1 is displayed on the right and H.264/AVC on the left, both limited to 100 kbps. The H.264/AVC video appears noticeably blurry, while the AV1 video remains much clearer — highlighting the significant advantage of AV1 for video calls under **bandwidth constraints.**